Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 15 Minutes

Instructor Solution - Quiz No 4

[20 points]

Note - *Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.*

Question 1 [20 points] Discrete Random Variables

We have a tetrahedral die (4-sided) that has four outcomes, each side bearing numbers 1, 2, 3 and 4 with equal probabilities (fair die). We define X_1 and X_2 as the first and second outcome of the two independent rolls of the tetrahedral die, respectively. We define the following three Discrete Random Variables (DRV's):

i. $\boldsymbol{Z_2 = \max(X_1\,;\,X_2)}$ - meaning Z_2 is the maximum of the outcomes of the two independent throws of the tetrahedral die. *Example if* $(X_1 = 2\ \&\ X_2 = 3)$, *then* $Z_2 = 3$.

ii. $\boldsymbol{Y_2 = \min(X_1\,;\,X_2)}$ - meaning Y_2 is the minimum of the outcomes of the two independent throws of the tetrahedral die. *Example if* $(X_1 = 2\ \&\ X_2 = 3)$, *then* $Y_2 = 2$.

iii. $\boldsymbol{W = Z_2 - Y_2}$.

(a) **[3 points]** Calculate and plot the Probability Mass Function, Pmf for DRV, Z_2.

(b) **[3 points]** Calculate and plot the Probability Mass Function, Pmf for DRV, Y_2.

(c) **[8 points]** Calculate and plot the Probability Mass Function, Pmf for DRV, W.

(d) **[6 points]** Calculate the Expected value, $E[W]$ and Variance of DRV, W.

X_1	X_2	Probabilities	Z_2	Y_2	W
1	1	1/16	1	1	0
1	2	1/16	2	1	1
1	3	1/16	3	1	2
1	4	1/16	4	1	3
2	1	1/16	2	1	1
2	2	1/16	2	2	0
2	3	1/16	3	2	1
2	4	1/16	4	2	2
3	1	1/16	3	1	2
3	2	1/16	3	2	1
3	3	1/16	3	3	0
3	4	1/16	4	3	1
4	1	1/16	4	1	3
4	2	1/16	4	2	2
4	3	1/16	4	3	1
4	4	1/16	4	4	0

Table 1

Z_2	Occurrences	Probability
1	1	1/16
2	3	3/16
3	5	5/16
4	7	7/16
Y_2		
1	7	7/16
2	5	5/16
3	3	3/16
4	1	1/16
W		
0	4	4/16
1	6	6/16
2	4	4/16
3	2	2/16

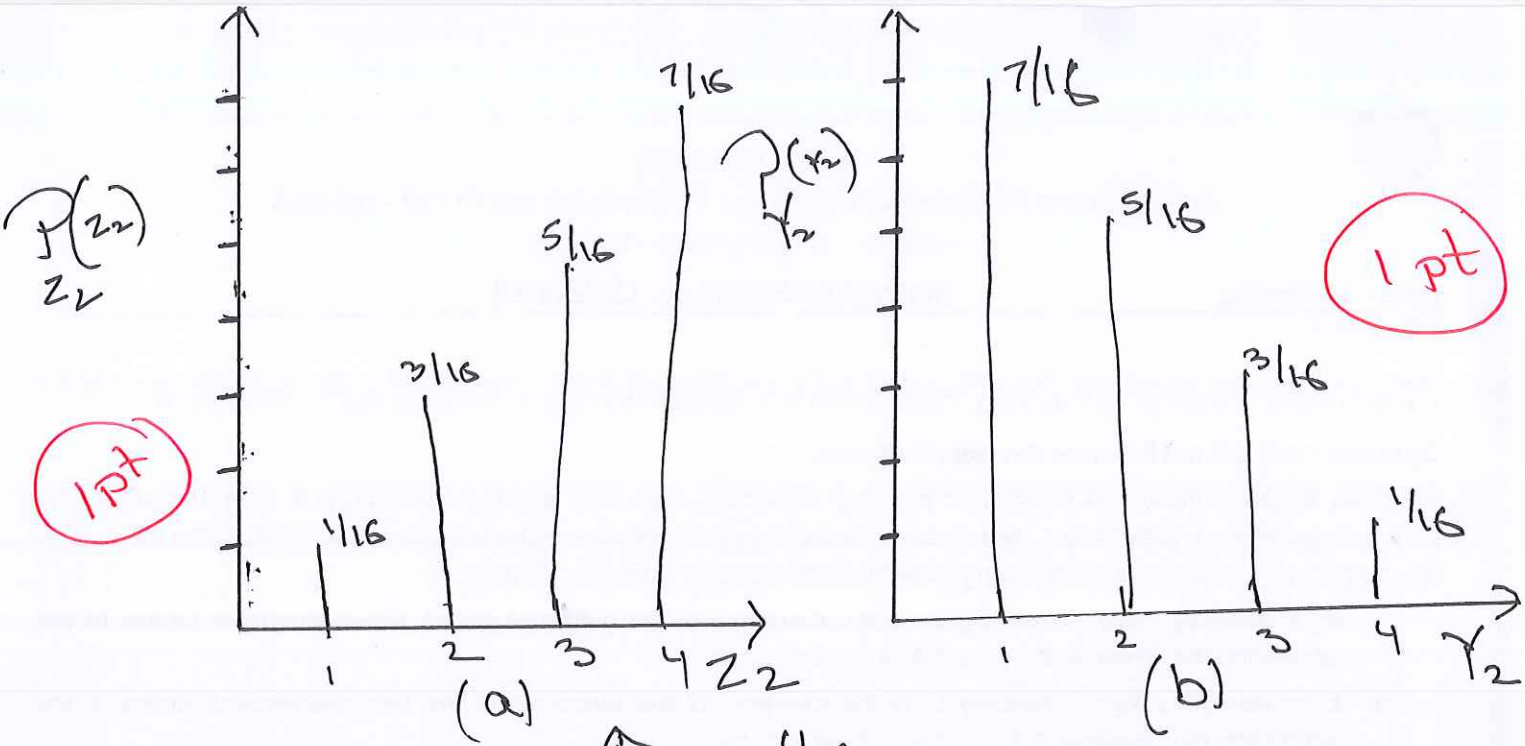
Table 2

2 pts

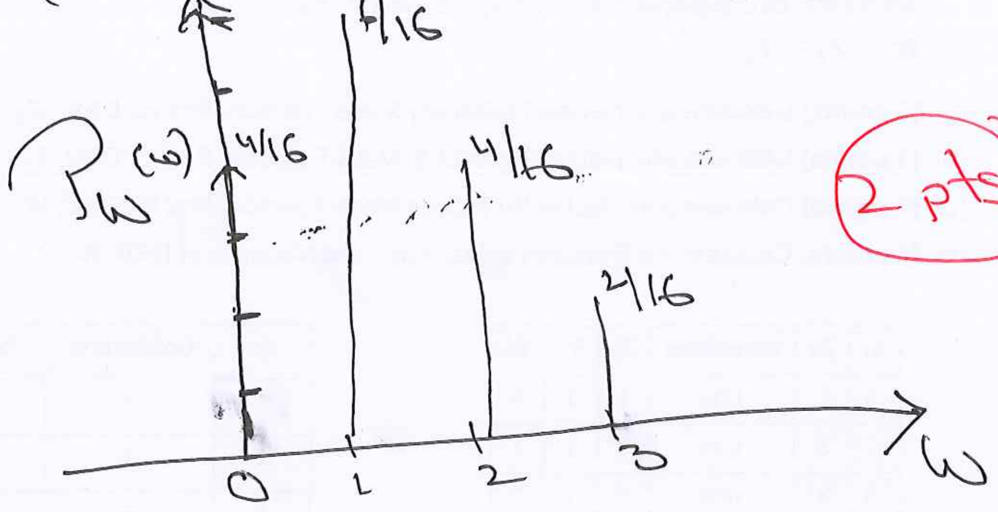
2 pts

Cal W +

6 pts



See Table 182



(d) $E(W) = 0 \cdot 4/16 + 1 \cdot \frac{6}{16} + 2 \times \frac{4}{16} + 3 \times \frac{2}{16}$ — 2pts

$$= \frac{1}{16}(6+8+6) = \frac{20}{16} = 1.25$$ — 1pt

$$Var(X) = E[X^2] - [E(X)]^2$$

$$= \left[1 \times \frac{6}{16} + 4 \times \frac{4}{16} + 9 \times \frac{2}{16}\right] - (1.25)^2$$ — 2pts

$$= \frac{1}{16}(6 + 16 + 18) - 1.25^2$$

$$= 2.5 - 1.5625$$

$$= 0.9375$$ — 1 pts